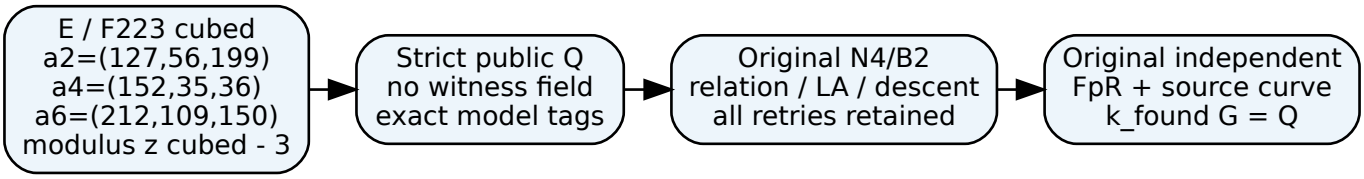
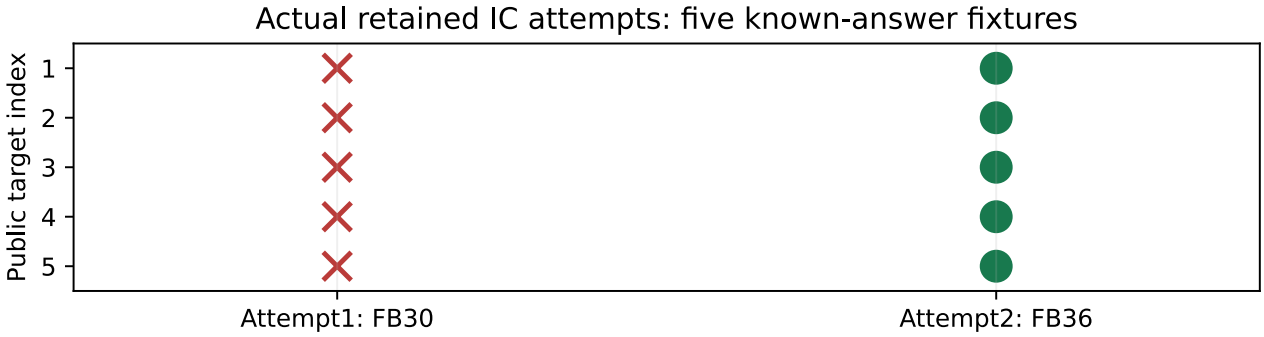
Public-target N4: native correctness

2026-11-13 · one fixed case / five known-answer targets · all standing unsigned





Observed, unsigned e08eac: three unfiltered native tests pass, zero failed/ignored/filtered, all2188 source guards unchanged. All five public IC targets solve through the literal original-body derivative; all five answers pass the original independent FpR/curve check. Ten paired legacy/public attempts agree on success/failure/recovered scalar. The five first attempts fail with26rows/30columns; the five second attempts with FB36 succeed. The chart records that complete attempt history, not a performance comparison.

Full source identity (identity.json is adjacent): q223, seed1, extension degree3, modulus [220,0,0,1]. G=((162,197,140),(45,11,207)); ell2773073, cofactor4, #E(Fq³)=11092292. Original independent irreducibility/primality/curve/nonzeroG/ellG checks and the unique Hasse multiple4 certify this source order. Hasse interval [11082908,11096228], radius6660. Native j=(166,136,127), outside Fq; this is the existing restricted GHS family, not a generic prime-field curve result.

Field tag: K3-223-d9c024d6fb0b129b7ea9d1914c9c6c98e75db9e090311fbc32d96f7f56160213

Model tag: N4-E0-quadratic-v1-ea39671464a106703d154b1b34299e40dff57bb1fc4305ddc877691cd28c2238

Executable, proofs and exact limits

Actual executable interface, 9e522f: describe and solve processes return0 for the same five literal public targets; the entire description JSON matches the unit fixture. All14 invalid-input processes return2: foreignq/ schema/field/model, noncanonical coordinate, off-curve point, actual nonmember, unknown scalar, unknown witness, empty/six targets, oversize, overwrite and extra argument. Rejected outputs are absent and the original overwrite bytes remain unchanged. Actual output retains all ten IC attempts. Known-answer witnesses stay outside public requests.

Reuse and scope: four whole original N4 files are byte exact. The public solver derivative has only four literal plumbing substitutions: two scalar arguments become Q, and two target-construction lines are removed. Original relation/LA/descent/random/verification bodies remain. Legacy code is test-only in the executable. Static compiler/own-source format checks pass with all2187 guards unchanged; original compiler warnings, including the unused test-helper parameter, are retained. No strict-Clippy or warning-free claim is made.

Partial formal result, f68241: the same nine basepoint/affine-log/model declarations pass both ORIGINAL producerLean4.35 and consumerLean4.34.1 gates, zero trusted declarations. All nine exact original PCC bindings locally pass, unsigned. Lean01 fails the final quartic proof body; wholefef40b is retained. Frozene58e94 precedes only the explicit proof-body rewrite after the Rust terminal guards. No theorem statement, verifier or trust manifest was weakened.

Written bridge and remaining work: a final divisor key shifts by delta when P0 changes, while the same raw degree-six input shifts by2delta. The common quartic ordinate reflection gives the proposed source orientation S=-I for the completed map; actual V2 field/model/P0/origin/fibre/norm checks remain open. The nine abstract/ring lemmas do not construct those objects or prove N4 relation-kernel correctness. The public rho derivative exists, but its old return type does not expose its recovered scalar; independent rho-answer replay remains a separate accessor task.

Custody and decision: complete static/unit/CLI trees are ONE standard tar/XZ each, sizes9005108/9008132/9010244 bytes (f426af/4db2f4/069a13), with every-member/raw/archive hashes and exact round trips. Genuine remote publication is a separate receipt. Continue to independent source CI, the rho accessor and additive V2 bridges. Parent norm/CI/raw results remain settled. Preserve the withdrawn subgroup-enumeration history, all compiler/launcher/proof and relation failures.

Limits: this is one fixed constructor and five known-answer correctness fixtures. No benchmark, timing ratio, computational advantage, complete Jacobian enumeration or full formal geometry is established. Original outside build cache is reused for the CLI and explicitly not a cold comparison. Future experiments require the original S3 checkpoint/live budget/capacity admission and strongest applicable references with all material cold costs.

Source-linked evidence

e08eac/unit-acceptance-01;9e522f/cli-acceptance-01;f68241/DERIVATION.md/PROOF_MAP.md/formal-scope-01;lean-bindings-01/claims.json;wholef426af/4db2f4/069a13;failedLean01fef40b/correctione58e94. Full native identity and editable figures accompany this report. Parent transfer graph remains unchanged; V2 edges are pending.